Admission Predictor + AI-Powered Graduate Guidance

By **Muhammad Baba Ibrahim**
FlexiSAF Intern 2026 - Generative AI and Data Science

Introduction

An AI-driven system that predicts graduate admission probability and provides personalized, actionable advice to applicants.

The solution combines a Random Forest model with a Generative AI counselor for clear, data-driven recommendations.

01

Project Overview and Objectives

Project Overview

Admission Success Predictor estimates the *Chance of Admit* using historical applicant data and returns tailored guidance via a GenAI agent.

Delivered as a Streamlit dashboard for interactive inputs and real-time feedback.

Objectives

Provide transparent admission probability, highlight key influencing factors, and offer personalized recommendations to improve application outcomes.

Enable scalable, AI-assisted counseling for students with limited advising resources.

Stakeholders

Primary: prospective graduate students seeking admission guidance.

Secondary: university advisors, career services, and internship program sponsors.

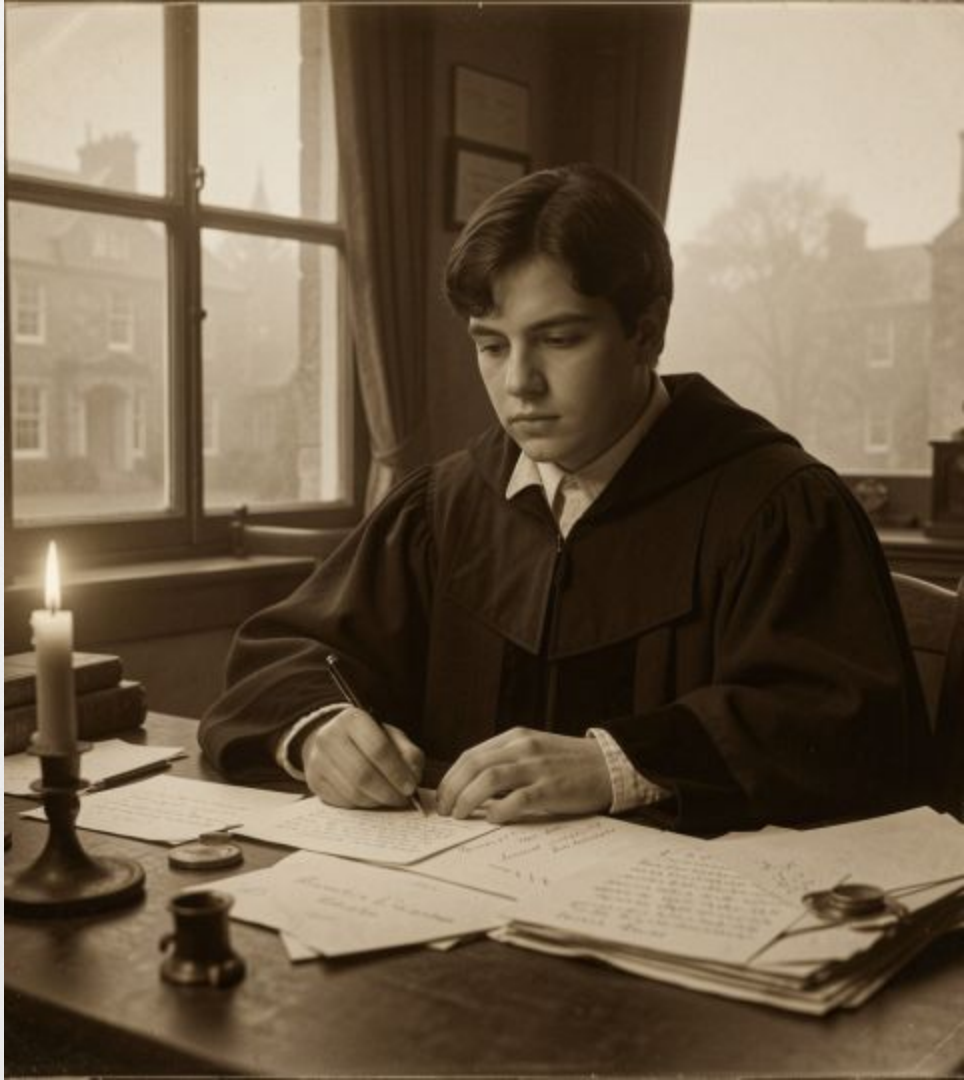
01

Problem Statement and Dataset Details

Problem Statement

Applying to graduate school is uncertain and often lacks personalized, data-driven advice.

Our system reduces uncertainty by predicting admission probability and recommending targeted actions to improve chances.



Dataset Details

Uses the Graduate Admission Dataset (Kaggle) with features: GRE, TOEFL, CGPA, University Rating, SOP, LOR, Research Experience.

Target: *Chance of Admit* (probability 0–1). Preprocessing includes scaling and encoding.

Target Variable

Chance of Admit is the target: a probability (0–1) representing admission likelihood.

Develop a reliable classifier to estimate Chance of Admit (0–1) and flag applicants as Admissible or At Risk using a 0.60 threshold.

Model must ensure consistent preprocessing, avoid data leakage, and report feature importances for transparency.

Predicted value guides personalized recommendations and decision-making for applicants.

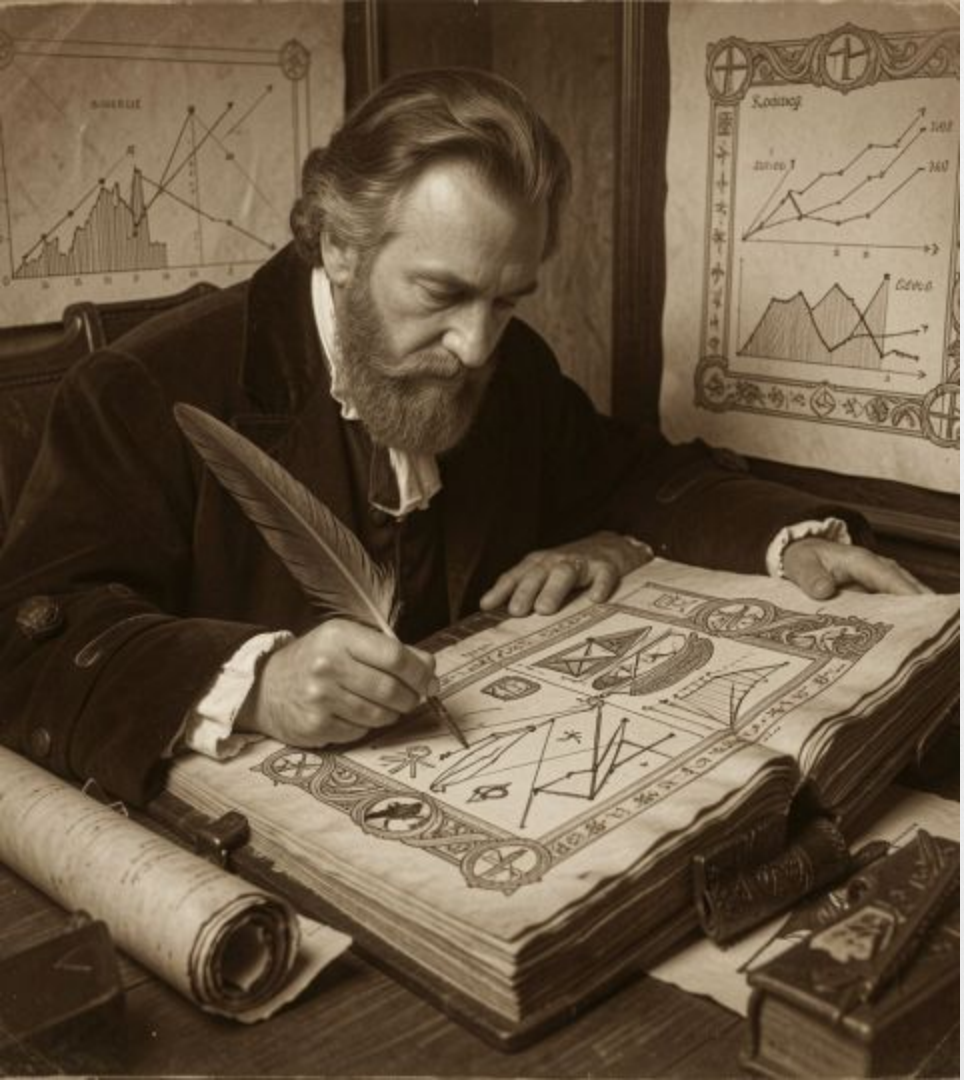
02

Model Training, AI Councilor, and Deployment

ML Pipeline

Preprocessing: scaling numeric features and one-hot encoding categorical variables.

Training: Random Forest classifier with 80/20 split, cross-validation, and joblib persistence for reproducible deployment.



GenAI Counselor

A LangGraph-managed agent uses model output and applicant data to generate natural-language guidance via Gemini.

Provides actionable suggestions (e.g., retake GRE, strengthen SOP) tailored to the predicted probability.

Deployment

Delivered as a Streamlit dashboard with real-time sliders and probability gauge.

Model served with a saved pipeline; GenAI integrated for live responses and easy institutional rollout.

02

Model Result and Visualization

Model Results

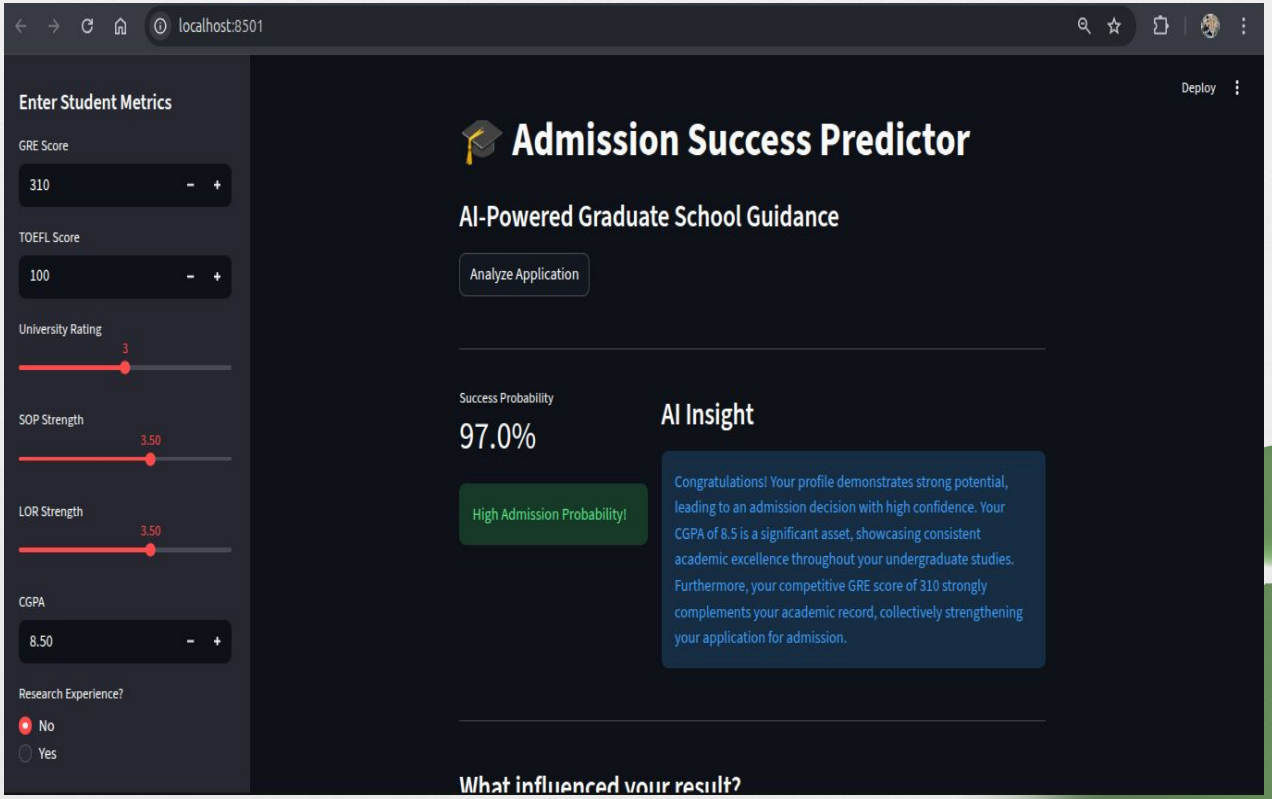
Random Forest achieved strong performance: ~90% accuracy, ROC-AUC ~0.94, F1 ~0.88.

Top drivers: *CGPA*, *GRE*, and research experience—used to explain predictions to users.

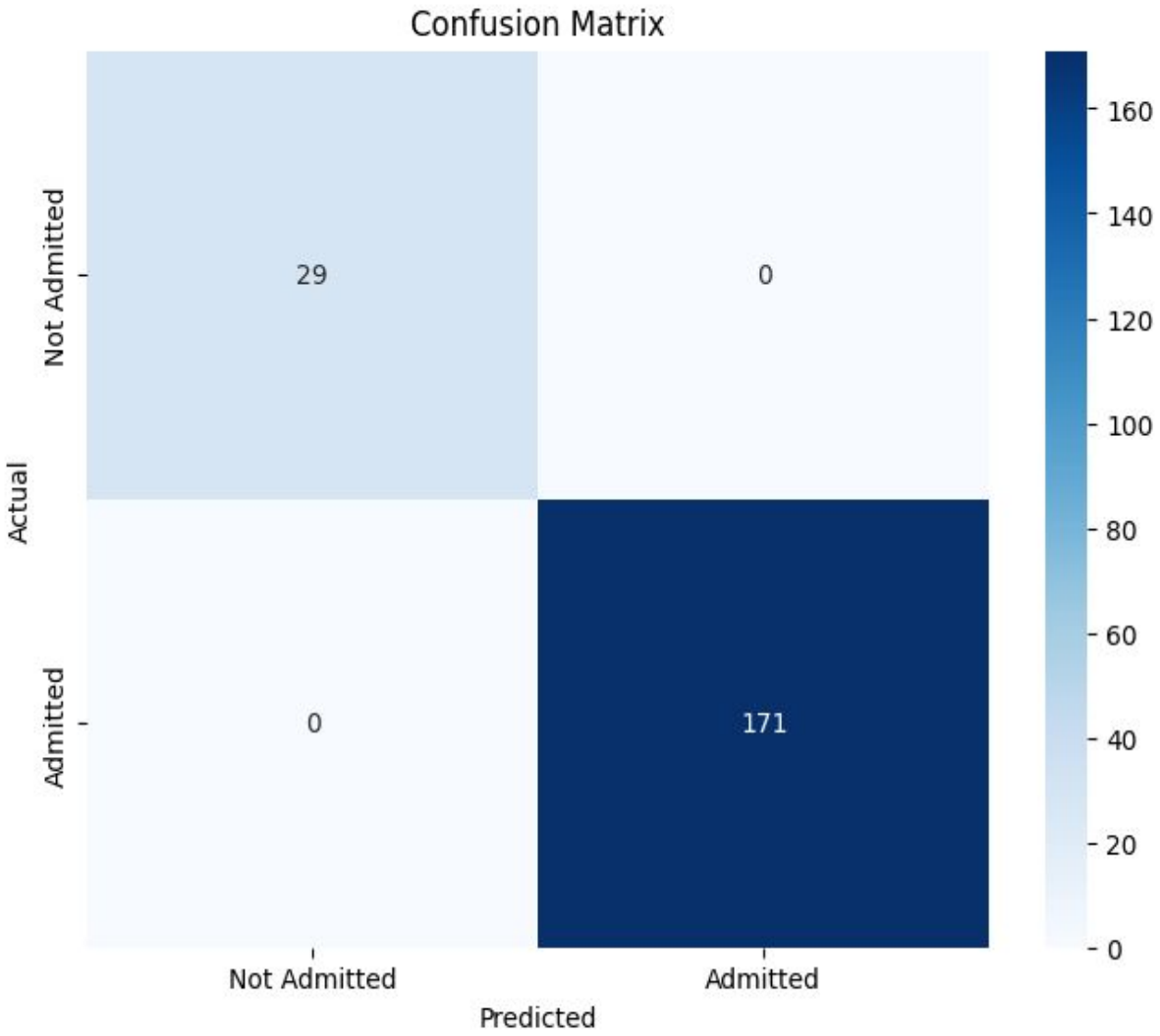
Visualizations

Interactive charts include feature importance, ROC curve, confusion matrix, and distribution plots.

Dashboard enables applicant comparison against average profiles for context and insight.



Confusion matrix to visualize the model's classification performance.



Future Work

Enhance model with additional features (letters, extracurriculars) and longitudinal outcomes.

Add A/B testing for GenAI messages and multi-language support for wider accessibility.

Conclusions

The Admission Predictor combines a robust Random Forest model and GenAI counselor to deliver transparent, personalized admission advice.

Scalable dashboard deployment supports improved decision-making and equitable access to data-driven guidance.

Thanks!

Does anyone have any questions?

ibrahimmuhammad271@gmail.com
+2347013013462
https://goldeno10.github.io/

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